CS 412 / 512
Introduction to Computer Graphics

Zhu Wang
Assistant Professor of Computer Science



COMPUTER SCIENCE

2D Transformation

2D Transformation













Linear Transformation

A transformation maps points to other points and vectors to other vectors In 2D

$$(x, y) \rightarrow (x', y')$$

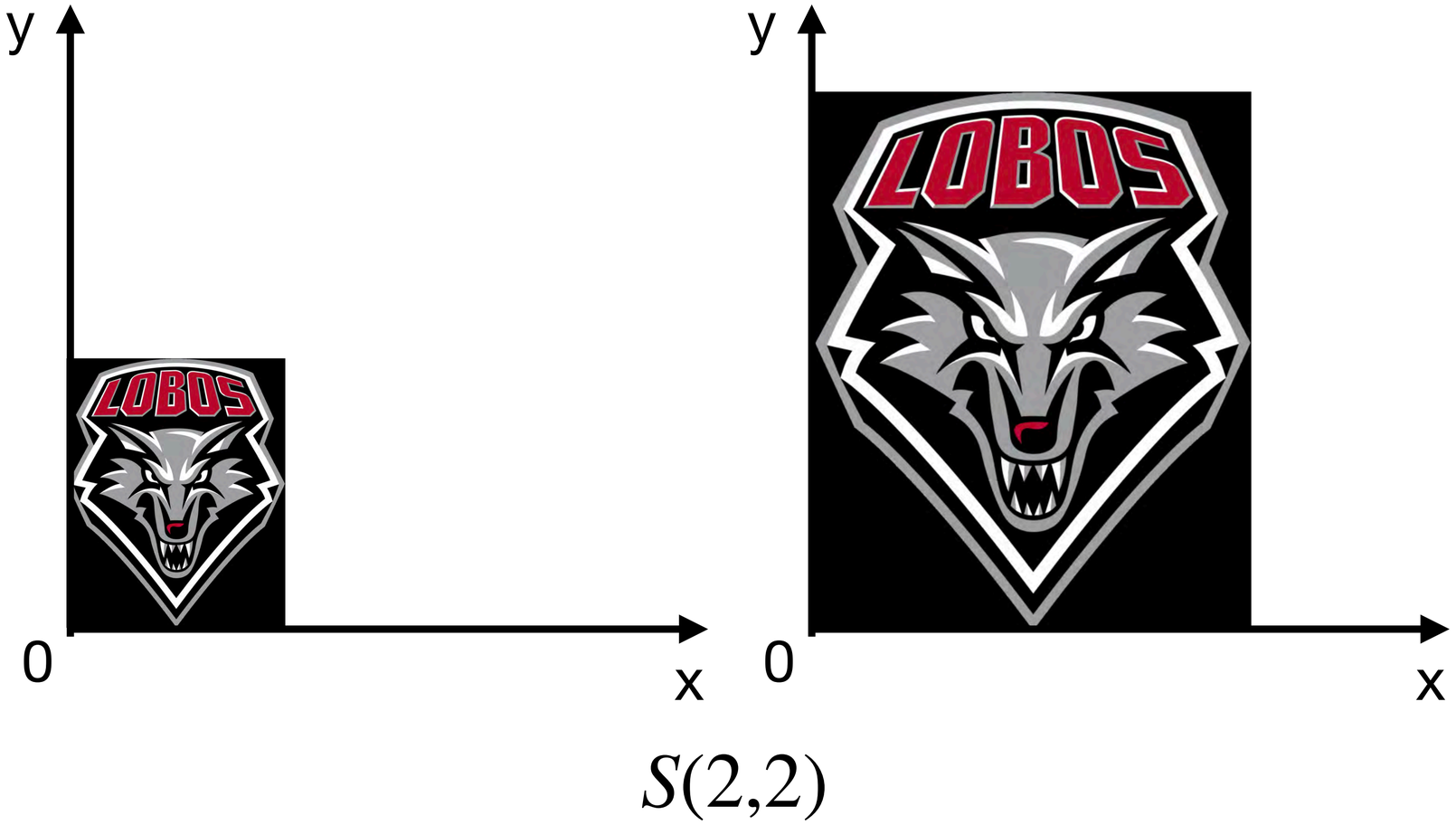
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Linear Transformation

2D Scaling

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$S = (s_x, s_y)$$



$S(2,2)$

Linear Transformation

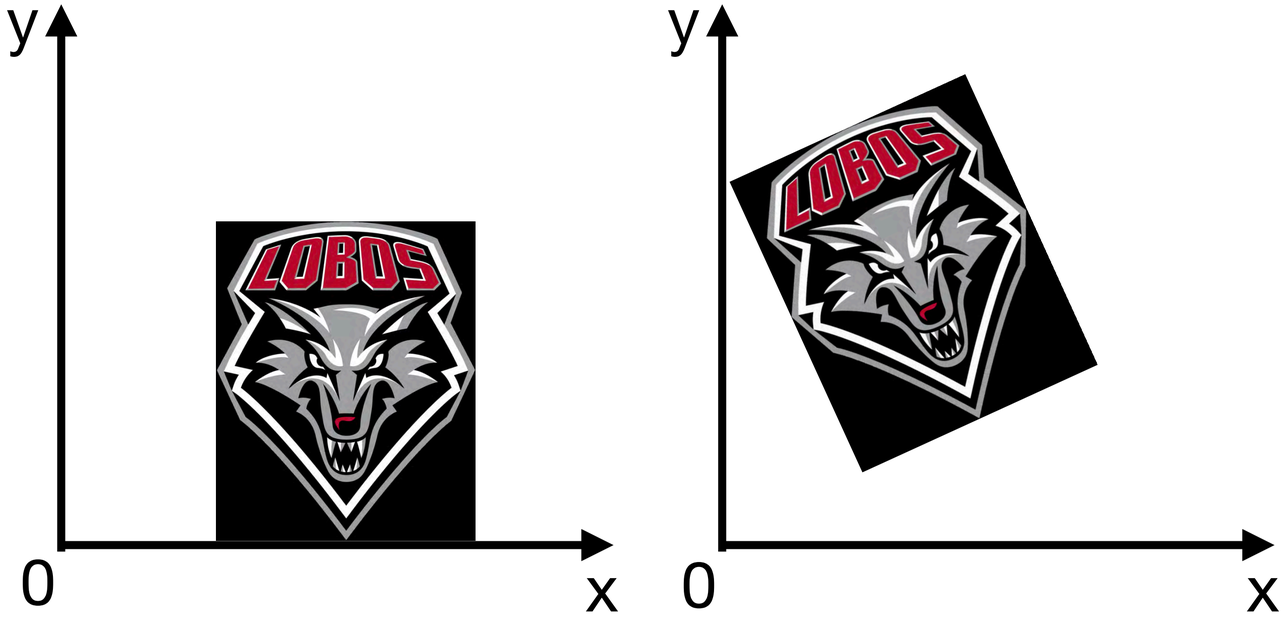
2D Rotation

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} cos\theta & -sin\theta \\ sin\theta & cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$Rot_{\theta}$$

$$Rot_{0^\circ} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

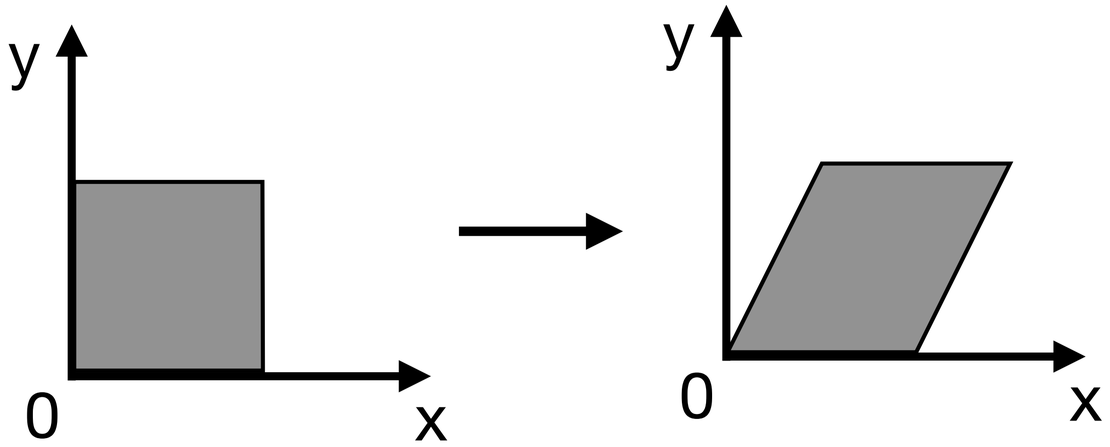
$$Rot_{90^\circ} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

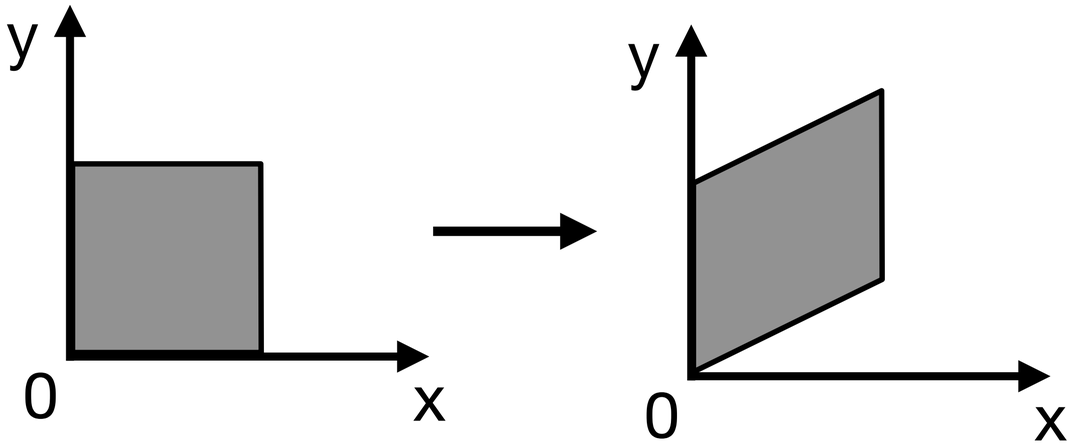


$$\theta = 30^\circ$$

Linear Transformation

Shearing/Skewing





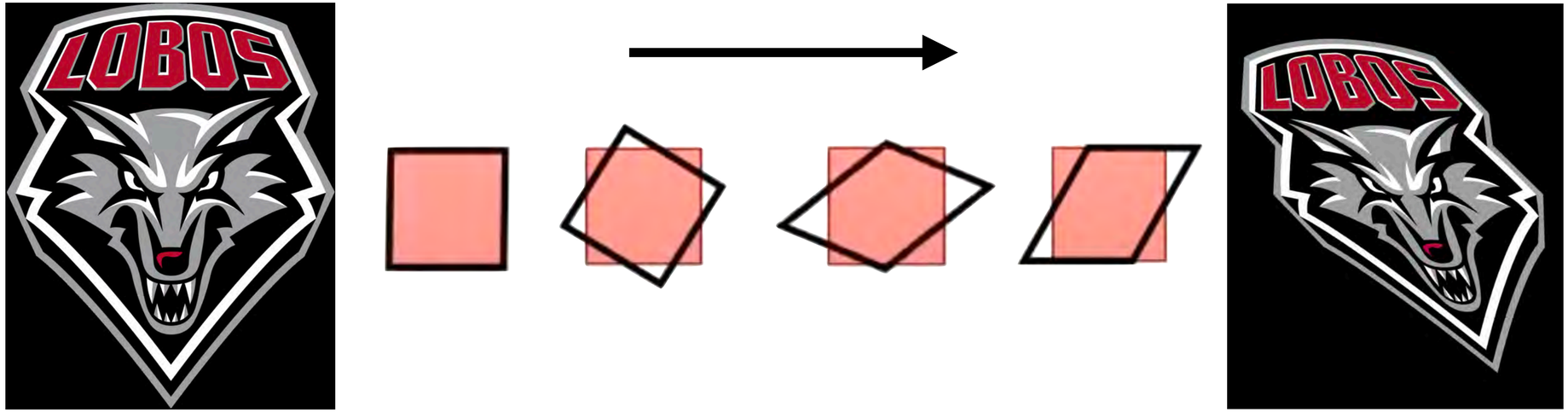
X-shear $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & sh_x \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

Y-shear $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ sh_y & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & sh_x \\ sh_y & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Linear Transformation

Shearing/Skewing



Linear Transformation

Mirroring/Reflecting



When a reflection is across the reflection line L, which is through the origin and makes an angle θ with the x-axis:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} cos2\theta & sin2\theta \\ sin2\theta & -cos2\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$Ref_\theta$$

Linear Transformation

Mirroring/Reflecting



$\theta = 0°$



$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$(x, y) \rightarrow (x, -y)$$

Reflection across the x-axis

$\theta = 90°$



$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

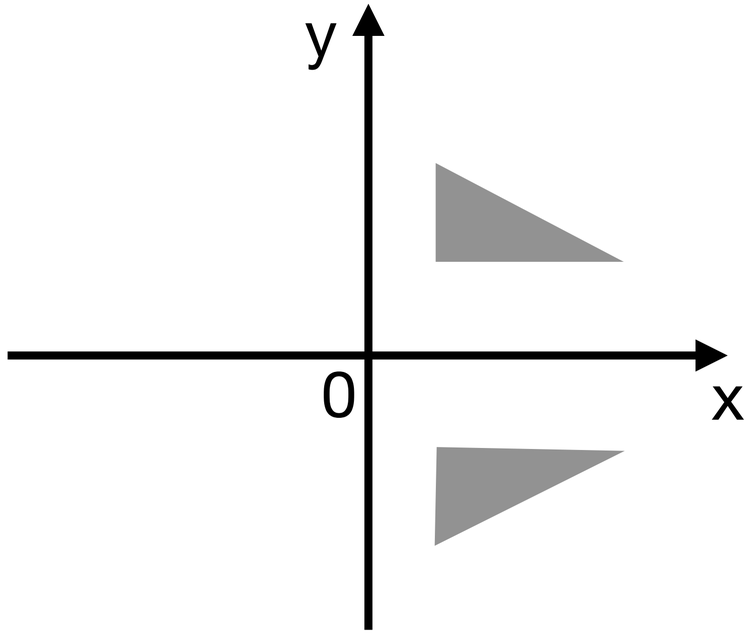
$$(x, y) \rightarrow (-x, y)$$

Reflection across the y-axis

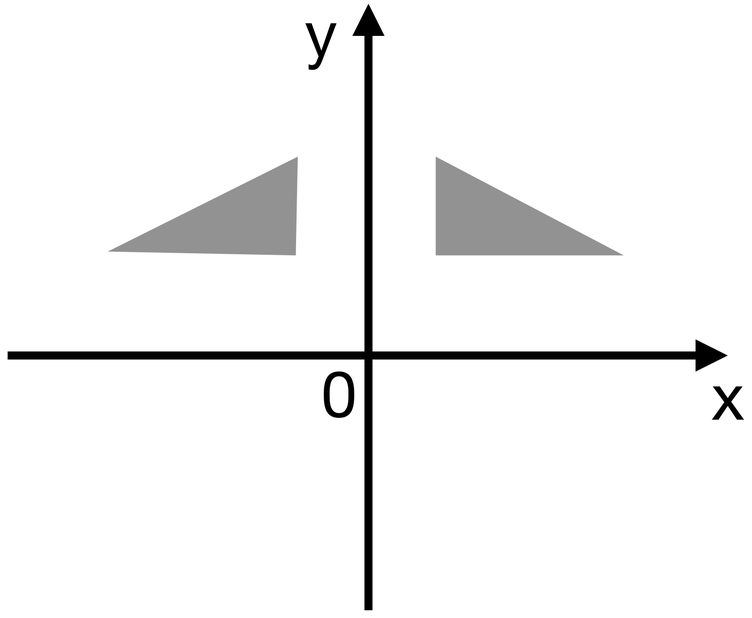
Linear Transformation

Mirroring/Reflecting

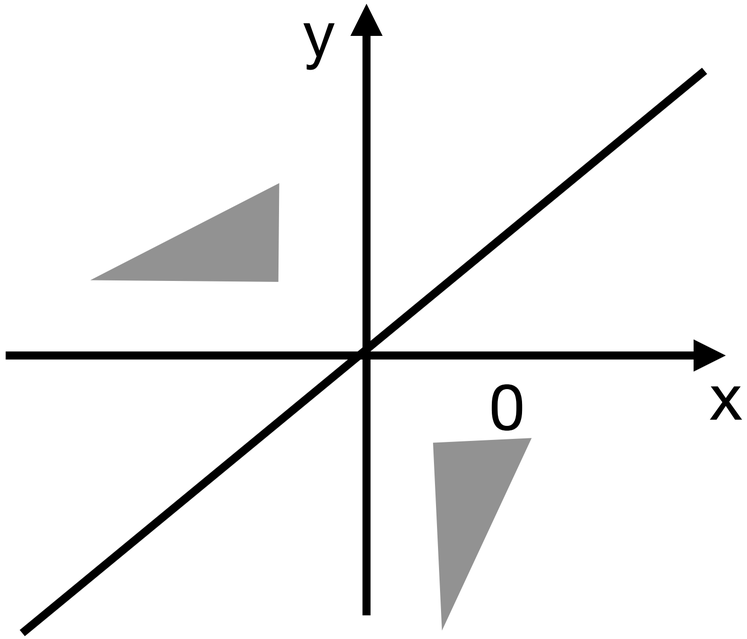
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} cos2\theta & sin2\theta \\ sin2\theta & -cos2\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$



$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$



$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$



$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Linear Transformation

Mirroring/Reflecting

When a reflection is across the reflection line L, which is through the origin and makes an angle θ with the x-axis:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} cos2\theta & sin2\theta \\ sin2\theta & -cos2\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$Ref_\theta$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix}$$

If L is not through the origin?

Translation, reflection, translation back. But how to represent translations in 2D matrix?

↓?

$$(x', y') = (x + c_1, y + c_2)$$

Affine Transformation

Translation & Homogeneous Coordinates

Affine map = Linear map + Translation

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix} \longrightarrow \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} a & b & t_x \\ c & d & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} ax + by + t_x \\ cx + dy + t_y \\ 1 \end{bmatrix}$$

2D point $(x, y) \rightarrow (x, y, 1)$

2D vector $[x\ y] \rightarrow [x\ y\ 0]^T$

Translation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x + t_x \\ y + t_y \\ 1 \end{bmatrix}$$

Homogeneous Coordinates

2D point $(x, y, 1)$

2D vector $[x\ y\ 0]^T$

point - point = vector

point + vector = point

vector + vector = vector

point + point → not meaningful in this context

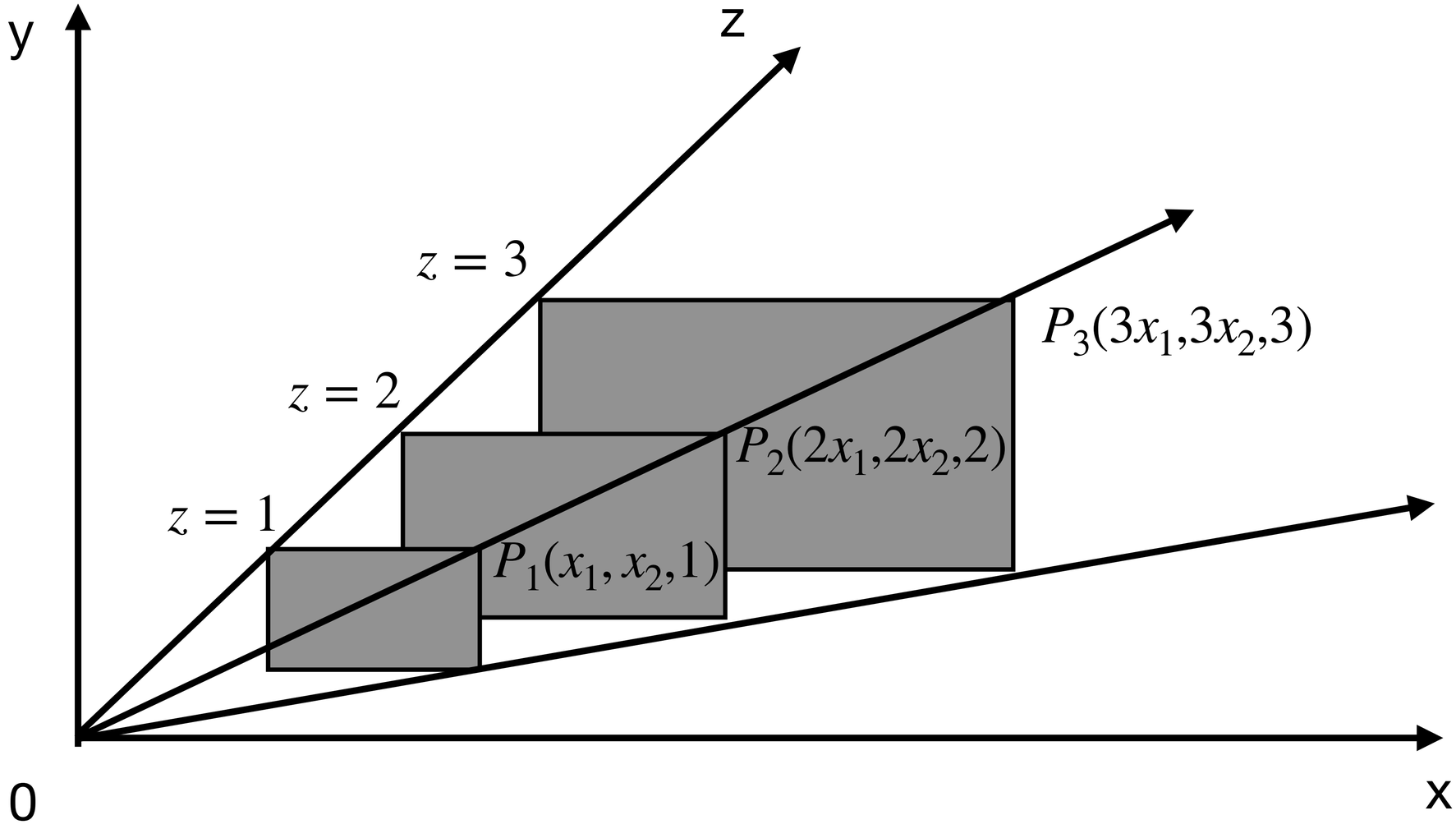
$$(x_1, y_1, 1) + (x_2, y_2, 1) = (x_1 + x_2, x_1 + y_2, 2)$$

$$\left(\frac{x_1 + x_2}{2}, \frac{x_1 + y_2}{2}, 1\right)$$

Geometric Interpolation

Think of it as interpolation between two hyperplanes

3D Representation of Homogeneous Space



Homogeneous Coordinates

Affine Transformation

Scaling

$$S(s_x, s_y) = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Rotation

$$Rot(\theta) = \begin{bmatrix} cos\theta & -sin\theta & 0 \\ sin\theta & cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Translation

$$T(t_x, t_y) = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

Transformation Concatenation

Affine Transformation

Matrix multiplication is not commutative $A \cdot B \neq B \cdot A$

If we want to have a sequence of affine maps $A_1, A_2, \cdots, A_n$

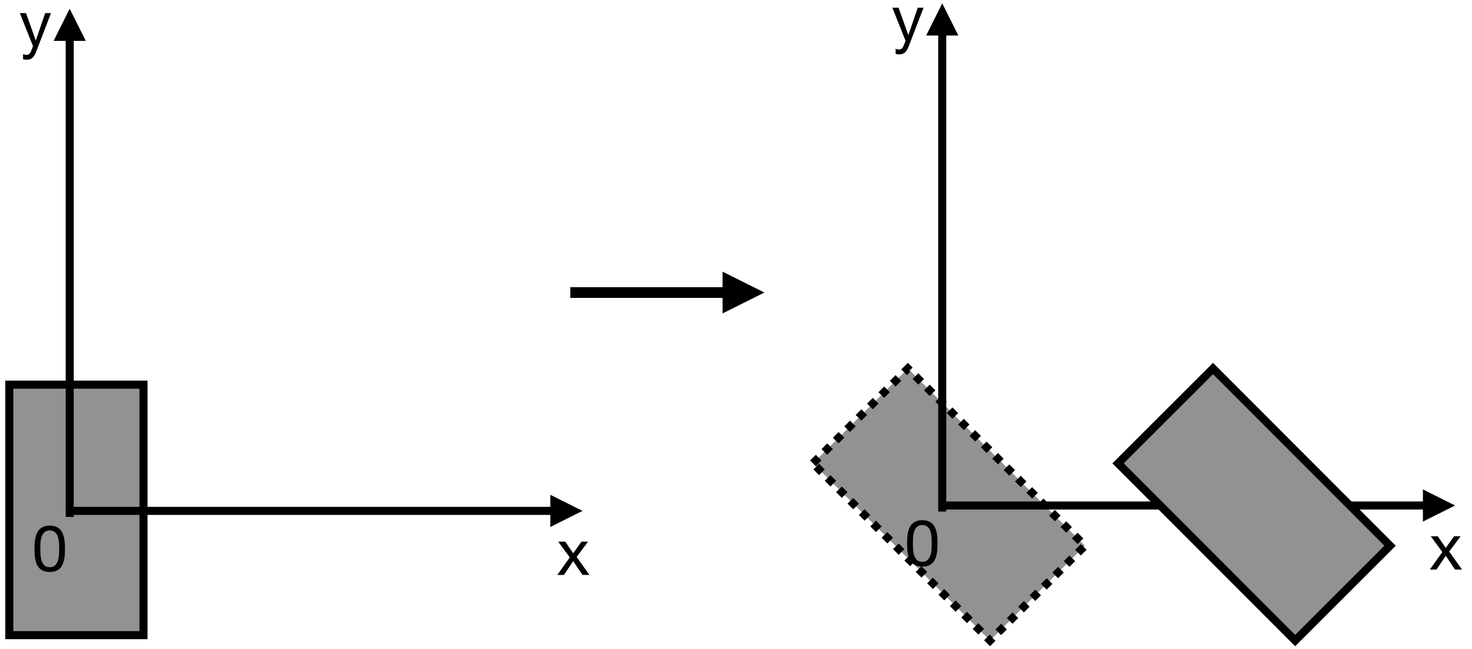
$$A_n(\cdots A_2(A_1 \cdot \mathbf{x}))) = A_n \cdots \cdot A_2 \cdot A_1 \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Transformation Order

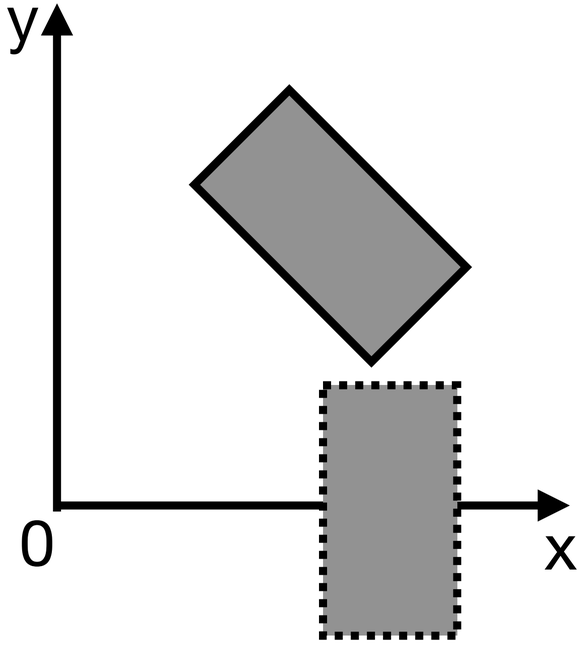
Translation, Rotation, Scaling

In general case, matrix multiplication is not commutative $A \cdot B \neq B \cdot A$

Rotation then translation



Translation then rotation

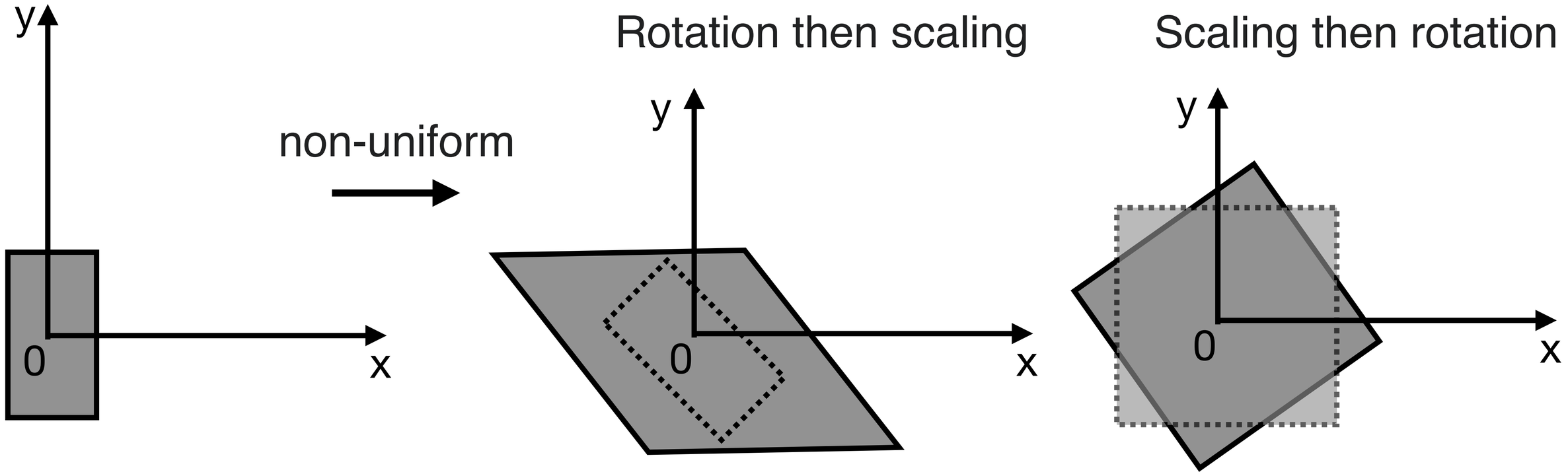


Transformation Order

Translation, Rotation, Scaling

When scaling is non-uniform $S \cdot R \neq R \cdot S$

When scaling is uniform $S \cdot R = R \cdot S$



Transformation Order

Translation, Rotation, Scaling

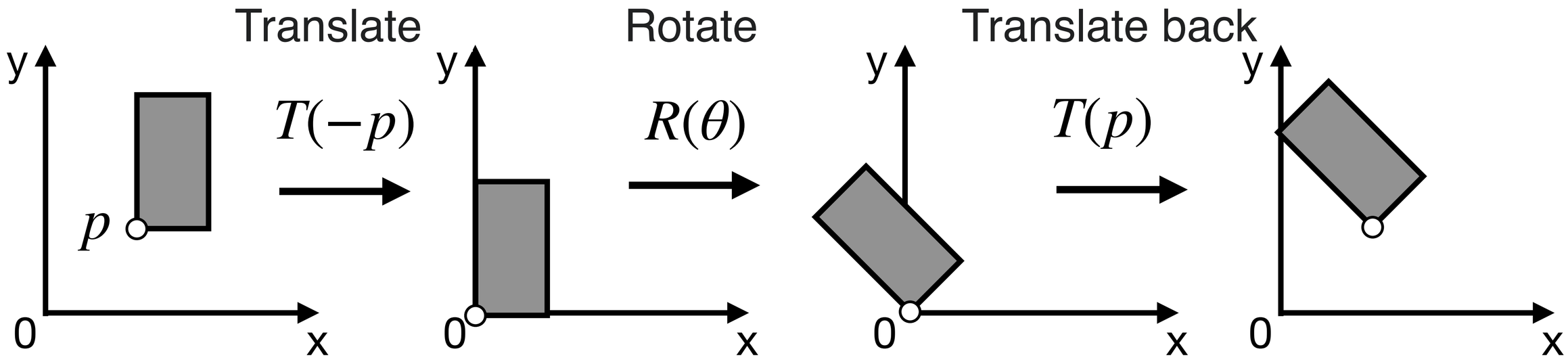
The most common pipeline

The final transformation matrix $M = T \cdot R \cdot S$

$$p' = M \cdot p = T \cdot R \cdot S \cdot p$$

Rotation Around an Arbitrary Point

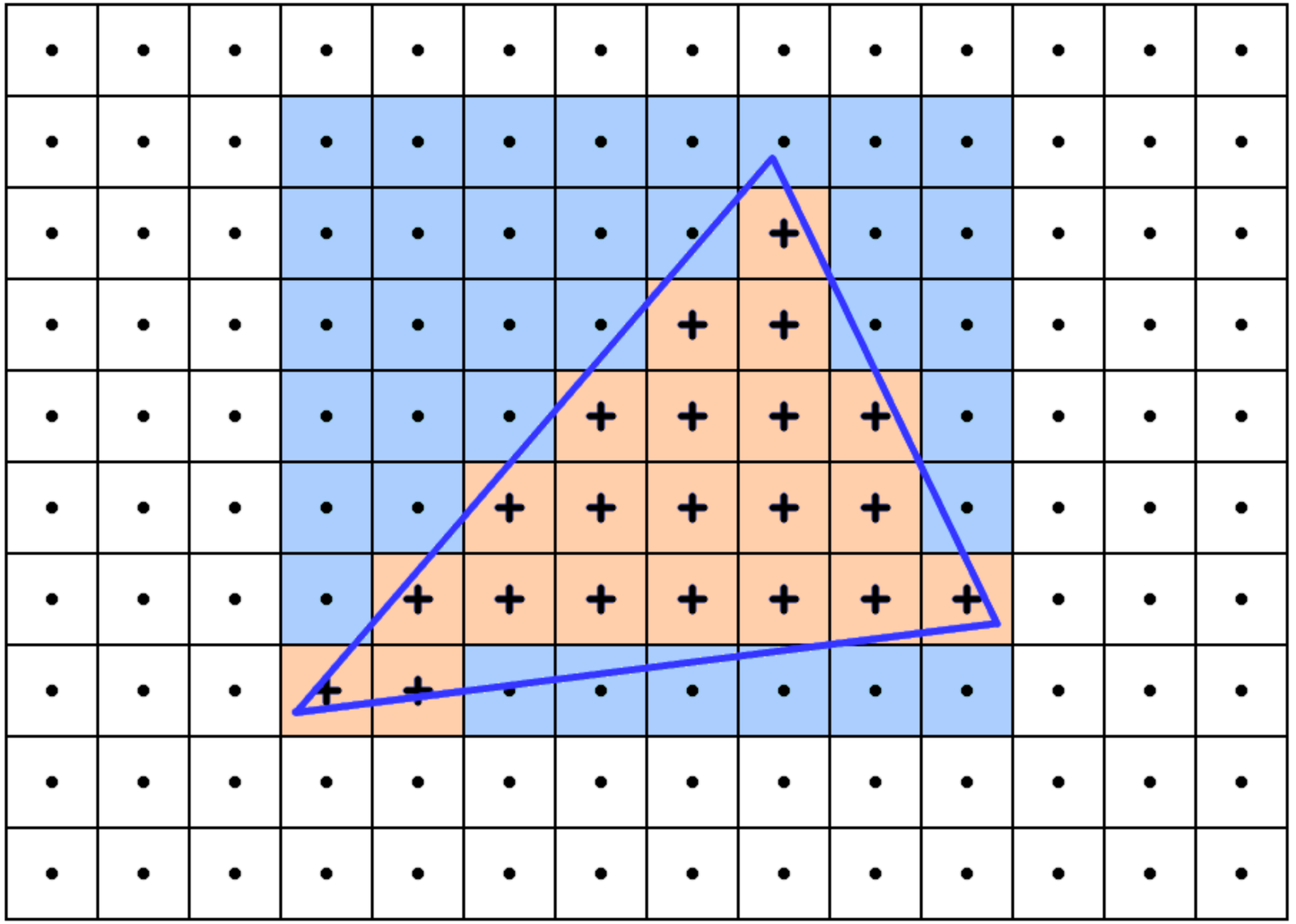
It is straightforward to rotate around origin, but rotation around p ?



$$T(p) \cdot R(\theta) \cdot T(-p)$$

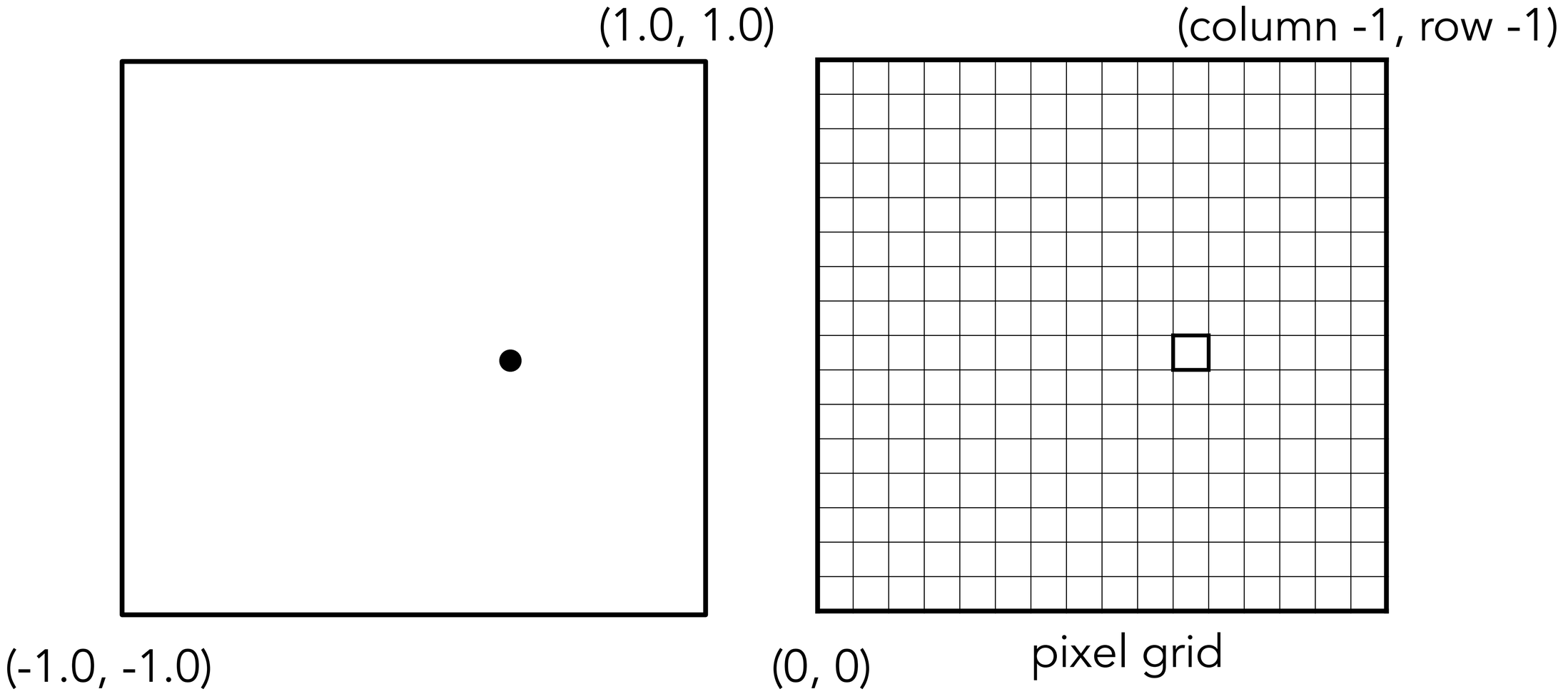
Rasterization

Rasterization

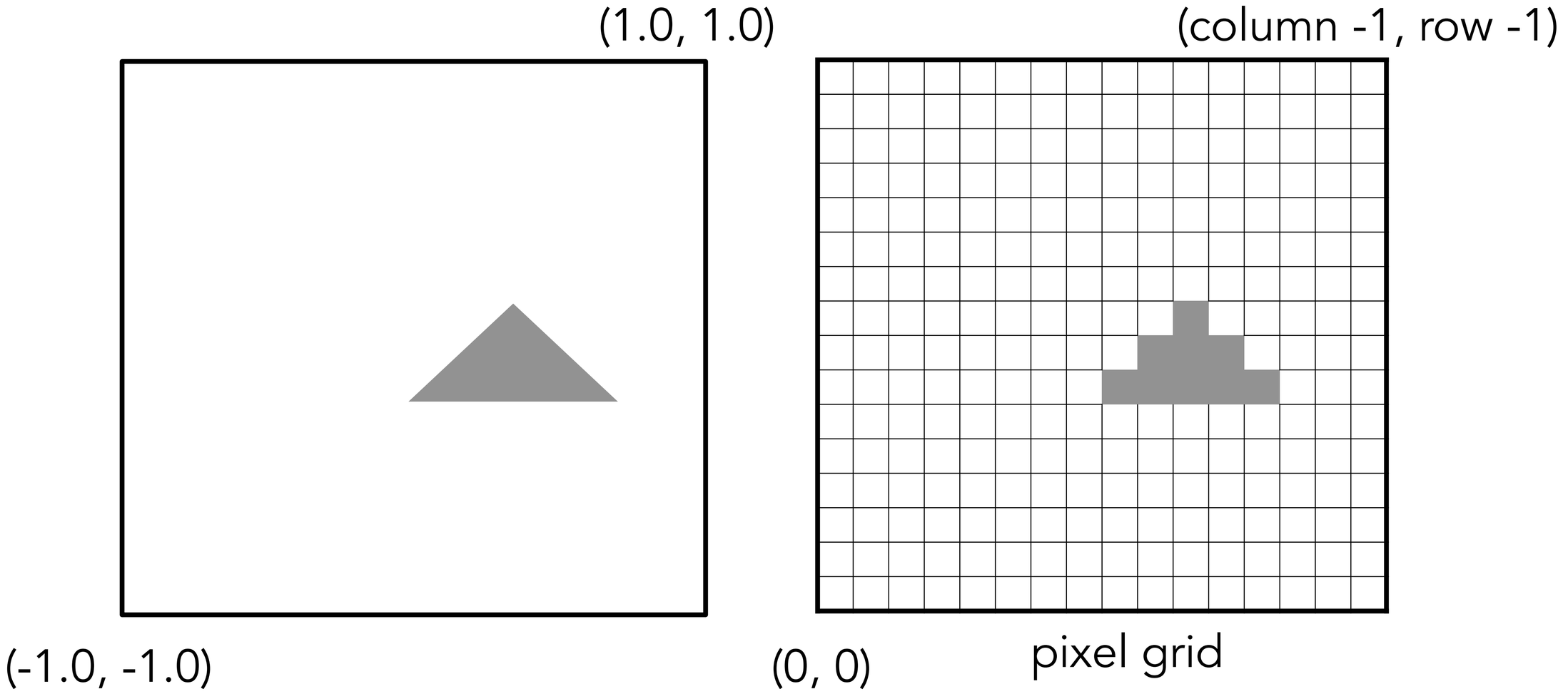


[source:Lisyarus Blog]

Rasterization

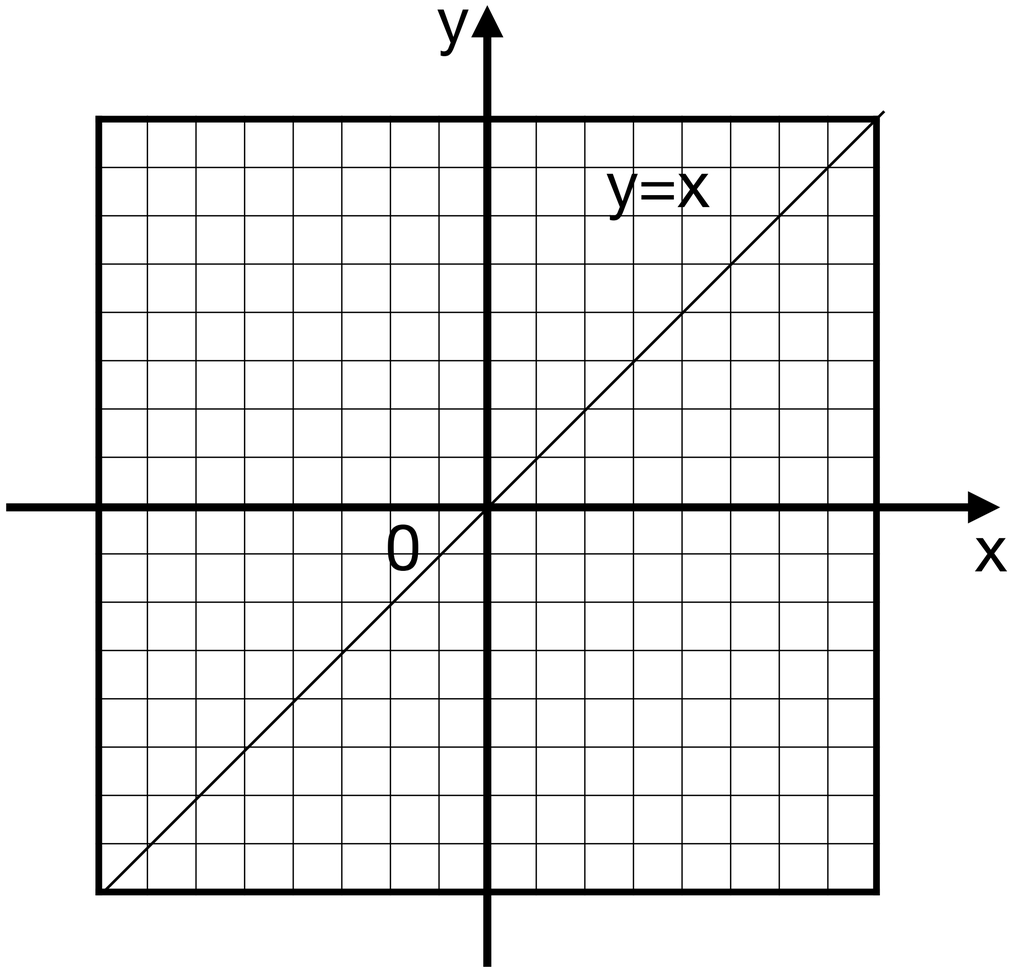


Rasterization



Explicit Representation

$y = f(x)$



For each x,

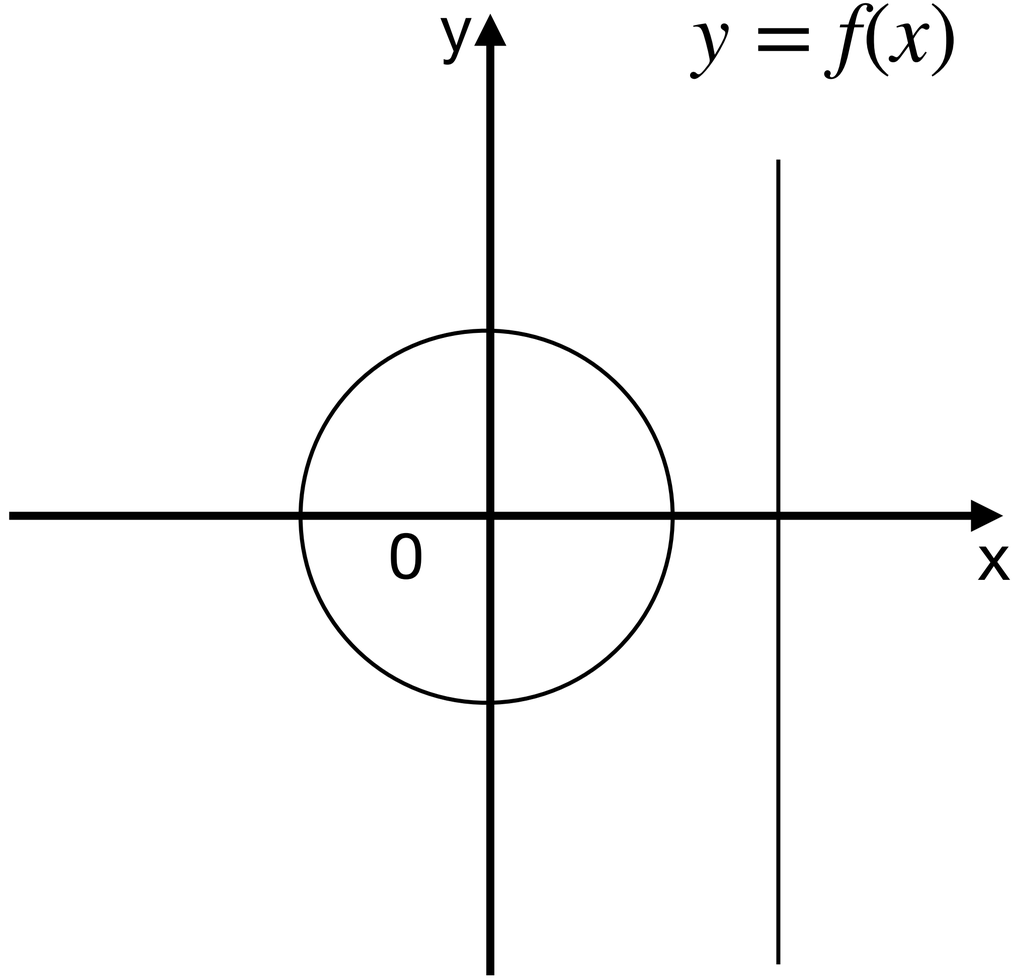
finding the pixels by $\lfloor y \rceil, \quad y = f(x)$

Explicit Representation

Limitations

?

$y = f(x)$



Implicit Representation

$$f(x, y) = 0$$

Lines $f(x, y) = ax + by + c$

Circles $f(x, y) = (x - c_x)^2 + (y - c_y)^2 - r^2$

On the shape $f(x, y) = 0$

Inside the shape $f(x, y) < 0$

Outside the shape $f(x, y) > 0$